ALGEBRA 2 · UNIT 2: LINEAR FUNCTIONS & SYSTEMS

Farm Planner

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan crops for Sunrise Farm. They graph constraint inequalities, find the vertices of the feasible region, maximize profit with linear programming, discover a tie when the budget line is parallel to the profit line, and solve a three-variable system to mix animal feed.

Linear programming shows why systems and inequalities matter to real decisions. The Honors budget constraint is parallel to the profit line, creating many optimal plans: a rich discussion.

Standards Addressed

- A-CED.3 Represent constraints by systems
- A-REI.6 Solve systems of linear equations
- A-REI.12 Graph systems of inequalities
- A-REI.C Systems in three variables

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: guess the best plan. Part 1 constraints and graph.
Day 2	Part 2 vertices and optimization.
Day 3	Part 3 feed mix and plan.

Materials

- Student handout
- Graph paper and rulers
- Graphing calculators or Desmos
- Optional: local farm data

How the Three Versions Differ

Level	What changes
CP2	Two constraints with a given graph, vertices by substitution, and profit at each vertex. Feed mix with two variables.
CP1	Writes constraints, finds vertices, optimizes, and solves a three-variable feed system.
Honors	Adds a budget constraint that creates a tie, sensitivity analysis, and a three-variable system with interpretation.

Watch For (Common Misconceptions)

- Testing points that are not vertices.
- Forgetting $x \geq 0$ and $y \geq 0$.
- Arithmetic errors in three-variable elimination.

Extension Ideas

- Add a fertilizer constraint.
- Use a spreadsheet solver.
- Interview a local farmer about planning.

Answer Key – CP2

Part 1: Constraints

#	Task	Answer
1	The farm has at most 200 acres. Write an inequality.	$x + y \leq 200$
2	Corn needs 2 hours of labor per acre and soybeans need 1 hour. There are at most 300 hours. Write an inequality.	$2x + y \leq 300$
3	Why must $x \geq 0$ and $y \geq 0$?	You can't plant a negative number of acres.
4	Is the plan (90, 100) possible? Is (120, 90)? Check both constraints.	(90, 100): $190 \leq 200$ and $280 \leq 300$, yes. (120, 90): $210 > 200$, no.

Part 2: Find the Best Plan

#	Task	Answer
5	Find the vertex where the land and labor lines meet by solving the system $x + y = 200$ and $2x + y = 300$.	Subtract: $x = 100$, $y = 100$
6	List all four vertices of the feasible region.	(0, 0), (0, 200), (100, 100), (150, 0)
7	Find the profit at each vertex. Which plan earns the most?	$0; $40,000; $50,000; $45,000. Best: 100 acres of each.

Part 3: Feed & Plan

#	Task	Answer
8	The farmer needs 100 lb of feed that costs $16 total. Write and solve a system for the pounds of corn and soy meal.	$c + s = 100$ and $0.10c + 0.30s = 16 \rightarrow 0.10c + 30 - 0.30c = 16 \rightarrow c = 70$ lb corn, $s = 30$ lb soy meal
9	Write a short business plan stating how many acres of each crop to plant and why.	Answers vary.
10	What would change if soybean profit rose to $350 per acre?	New profits: (0, 200) = $70,000; (100, 100) = $65,000. Plant all soybeans.

Answer Key – CP1

Part 1: Constraints

#	Task	Answer
1	Write all constraints and graph the feasible region.	$x + y \le 200$, $2x + y \le 300$, $x \ge 0$, $y \ge 0$
2	Find all vertices algebraically.	(0, 0), (0, 200), (100, 100), (150, 0)
3	Explain why the labor constraint makes some land-only plans impossible.	A plan with more than 150 acres of corn needs more than 300 hours of labor.

Part 2: Optimize

#	Task	Answer
4	Find the maximum profit and the plan that gives it.	$50,000 at (100, 100)
5	If corn profit drops to $150 per acre, what is the new best plan?	$P = 150x + 200y$: (0, 200) = $40,000, (100, 100) = $35,000, (150, 0) = $22,500. Plant 200 acres of soybeans.
6	The farmer adds a rule: at least 40 acres of corn. Add it to the graph. Does the best plan change?	$x \ge 40$ removes (0, 0) and (0, 200), adding (40, 0) and (40, 160). (100, 100) is still feasible and still best.

Part 3: Mix the Feed

#	Task	Answer
7	A 100-lb batch must have 16.6 lb of protein and cost $15. Protein: corn 9%, soy 44%, oats 12%. Cost per lb: $0.10, $0.30, $0.15. Write the system.	$a + b + c = 100$; $0.09a + 0.44b + 0.12c = 16.6$; $0.10a + 0.30b + 0.15c = 15$
8	Solve the system by elimination and check.	$a = 60$, $b = 20$, $c = 20$. Check: $5.4 + 8.8 + 2.4 = 16.6$; $6 + 6 + 3 = 15$.
9	Write a business plan with the crop plan and the feed recipe.	Answers vary.

Answer Key – Honors

Part 1: Constraints & Vertices

#	Task	Answer
1	Write all constraints and graph them.	$x + y \le 200$; $2x + y \le 300$; $150x + 100y \le 24{,}000$ ($3x + 2y \le 480$); $x, y \ge 0$
2	Find every vertex of the new feasible region.	(0, 0), (0, 200), (80, 120), (120, 60), (150, 0)
3	Which constraints are binding at each vertex?	(0, 200): land; (80, 120): land and budget; (120, 60): budget and labor; (150, 0): labor

Part 2: Optimize & Analyze

#	Task	Answer
4	Evaluate P at each vertex. What do you notice?	\$0; \$40,000; \$48,000; \$48,000; \$45,000. Two vertices tie.
5	Explain the tie using slopes. List three different optimal plans.	$P = 100(3x + 2y)$, so the profit line is parallel to the budget line $3x + 2y = 480$. Every point on the segment from (80, 120) to (120, 60) earns \$48,000, such as (100, 90).
6	For what corn profit k (with soybeans still \$200) would (120, 60) be the only best plan?	Profit $kx + 200y$ with slope $-k \div 200$ must be steeper than the budget (-1.5) and less steep than labor (-2): $300 < k < 400$.

Part 3: Mix the Feed

#	Task	Answer
7	Solve: $a + b + c = 100$; $0.09a + 0.44b + 0.12c = 16.6$; $0.10a + 0.30b + 0.15c = 15$.	$a = 60$, $b = 20$, $c = 20$
8	If the protein target changes to p lb (cost and total stay the same), write a and b in terms of p (with c also in terms of p).	From the cost and total equations: $c = 100 - 4b$ and $a = 3b$. Protein: $0.27b + 0.44b + 12 - 0.48b = p \rightarrow b = (p - 12) \div 0.23$, $a = 3(p - 12) \div 0.23$, $c = 100 - 4(p - 12) \div 0.23$. At $p = 16.6$: (60, 20, 20) ✓
9	Find the range of p that keeps every amount nonnegative.	$b \ge 0$ needs $p \ge 12$; $c \ge 0$ needs $b \le 25$, so $p \le 17.75$. Range: $12 \le p \le 17.75$.

Scoring note: Accept any correct parametric form that returns (60, 20, 20) at $p = 16.6$.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked. Goes beyond the minimum with a creative, realistic choice.	Recommendation is the true optimum (or a tie is explained), and the feed system is solved and checked.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Polished, organized, and visually engaging.	Business plan includes the constraints, graph, vertices, optimal plan, and the feed mix solution. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.